

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		<u>electrons</u> in an atom are excited and the additional energy promotes them to <u>higher energy levels</u> (1) (when the source of energy is removed) the electrons <u>fall back down</u> to a lower energy level <u>emitting energy/light</u> (in the form of a photon) (1)	2			2		
		(ii)		award (1) for either of following absorption spectrum comprises dark lines while emission spectrum comprises coloured lines absorption spectrum has coloured background while emission spectrum has dark background	1			1		
	(b)			$f = \frac{3.00 \times 10^8}{95 \times 10^{-9}}$ (1) $f = 3.16 \times 10^{15} \text{ Hz} = 3.16 \times 10^9 \text{ MHz}$ (1)		2		2	2	
	(c)	(i)		exothermic since percentage product decreases as temperature increases (1) system opposes increase in temperature and takes in heat by shifting equilibrium to the left /endothermic direction (1)			2	2		
		(ii)		decrease in moles since percentage product increases as pressure increases(1) system opposes increase in pressure by shifting equilibrium to the right / to the side of fewer moles of gas (1)			2	2		